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Overview of materials for Polyphthalamide (PPA), Mineral Filled

Categories: Polymer; Thermoplastic; Polyphthalamide (PPA); Polyphthalamide (PPA), Mineral Filled

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Polyphthalamide, Mineral Filled". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

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Physical Properties	Original Value	Comments
Density	1.28 - 1.80 g/cc	Average value: 1.51 g/cc Grade Count:17
Water Absorption	0.100 - 0.200 %	Average value: 0.134 % Grade Count:7
Moisture Absorption at Equilibrium	0.160 - 1.40 %	Average value: 0.424 % Grade Count:5
Linear Mold Shrinkage	0.00300 - 0.0150 cm/cm	Average value: 0.00848 cm/cm Grade Count:12
Linear Mold Shrinkage, Transverse	0.00450 - 0.0110 cm/cm	Average value: 0.00853 cm/cm Grade Count:9

Mechanical Properties	Original Value	Comments
Hardness, Rockwell R	125 - 126	Average value: 125 Grade Count:6
Tensile Strength, Ultimate	70.0 - 130 MPa	Average value: 100 MPa Grade Count:12
	24.1 - 115 MPa @Temperature 60.0 - 175 °C	Average value: 63.2 MPa Grade Count:2
Tensile Strength, Yield	71.0 - 174 MPa	Average value: 111 MPa Grade Count:5
	20.0 - 97.0 MPa @Temperature 100 - 175 °C	Average value: 46.2 MPa Grade Count:2
Elongation at Break	1.20 - 4.00 %	Average value: 1.78 % Grade Count:15
	1.70 - 15.0 % @Temperature 60.0 - 175 °C	Average value: 6.52 % Grade Count:4
Elongation at Yield	1.80 - 2.50 %	Average value: 2.15 % Grade Count:4
Modulus of Elasticity	4.40 - 11.6 GPa	Average value: 8.40 GPa Grade Count:15
	0.600 - 8.70 GPa @Temperature 60.0 - 175 °C	Average value: 3.81 GPa Grade Count:4
Flexural Yield Strength	110 - 263 MPa	Average value: 177 MPa Grade Count:15
	15.0 - 155 MPa @Temperature 100 - 175 °C	Average value: 59.6 MPa Grade Count:3
Flexural Modulus	3.51 - 10.8 GPa	Average value: 6.89 GPa Grade Count:15
	0.600 - 6.80 GPa @Temperature 100 - 175 °C	Average value: 2.69 GPa Grade Count:3
Compressive Yield Strength	152 - 185 MPa	Average value: 166 MPa Grade Count:3
Poissons Ratio	0.290 - 0.380	Average value: 0.318 Grade Count:5
Shear Modulus	2.20 - 4.20 GPa	Average value: 3.23 GPa Grade Count:5

Shear Strength	69.0 - 96.0 MPa	Average value: 85.7 MPa Grade Count:8
Izod Impact, Notched	0.310 - 5.30 J/cm	Average value: 0.826 J/cm Grade Count:12
Izod Impact, Unnotched	3.47 - 7.47 J/cm	Average value: 5.92 J/cm Grade Count:5
Izod Impact, Notched (ISO)	2.90 - 95.0 kJ/m²	Average value: 19.6 kJ/m² Grade Count:6
	4.00 - 4.00 kJ/m² @Temperature -30.0 - -30.0 °C	Average value: 4.00 kJ/m² Grade Count:1
Izod Impact, Unnotched (ISO)	31.0 kJ/m² - NB	Average value: 42.0 kJ/m² Grade Count:3
	23.0 - 23.0 kJ/m² @Temperature -5.00 - -5.00 °C	Average value: 23.0 kJ/m² Grade Count:1
Charpy Impact Unnotched	1.00 - 10.5 J/cm²	Average value: 4.38 J/cm² Grade Count:6
	8.00 - 8.00 J/cm² @Temperature -40.0 - -40.0 °C	Average value: 8.00 J/cm² Grade Count:1
Charpy Impact, Notched	0.200 - 0.800 J/cm²	Average value: 0.400 J/cm² Grade Count:6
	0.500 - 0.500 J/cm² @Temperature -40.0 - -40.0 °C	Average value: 0.500 J/cm² Grade Count:1

Electrical Properties	Original Value	Comments
Electrical Resistivity	1.00e+15 - 1.00e+16 ohm-cm	Average value: 6.14e+15 ohm-cm Grade Count:6
Dielectric Constant	3.90 - 4.70	Average value: 4.22 Grade Count:6
Dissipation Factor	0.00500 - 0.0220	Average value: 0.0123 Grade Count:6
Arc Resistance	120 - 125 sec	Average value: 124 sec Grade Count:4
Comparative Tracking Index	550 V	Average value: 550 V Grade Count:7

Thermal Properties	Original Value	Comments
CTE, linear	23.0 - 41.0 µm/m-°C	Average value: 31.8 µm/m-°C Grade Count:4
	17.0 - 95.0 µm/m-°C @Temperature 50.0 - 250 °C	Average value: 47.2 µm/m-°C Grade Count:5
CTE, linear, Transverse to Flow	40.0 - 80.0 µm/m-°C	Average value: 53.8 µm/m-°C Grade Count:4
	63.0 - 150 µm/m-°C @Temperature 50.0 - 250 °C	Average value: 109 µm/m-°C Grade Count:4
Melting Point	265 - 324 °C	Average value: 308 °C Grade Count:10
Maximum Service Temperature, Air	130 - 150 °C	Average value: 140 °C Grade Count:3
Deflection Temperature at 0.46 MPa (66 psi)	158 - 306 °C	Average value: 265 °C Grade Count:9
Deflection Temperature at 1.8 MPa (264 psi)	80.0 - 279 °C	Average value: 184 °C Grade Count:12
Flammability, UL94	HB	Grade Count:8

Processing Properties	Original Value	Comments
Melt Temperature	300 - 343 °C	Average value: 323 °C Grade Count:11
Mold Temperature	66.0 - 180 °C	Average value: 131 °C Grade Count:11
Drying Temperature	80.0 - 130 °C	Average value: 111 °C Grade Count:10
Moisture Content	0.0300 - 0.200 %	Average value: 0.0693 % Grade Count:7
Dew Point	-31.7 °C	Average value: -31.7 °C Grade Count:3
Injection Pressure	68.9 - 124 MPa	Average value: 96.5 MPa Grade Count:3

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